

Engineering Innovation & Entrepreneurship

03/29/2026

Everyday Friction

Profile Matrix

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1. Competitive Analysis Table

Key Success Factor	Weight	Everyday Friction		Vintar - Universal Travel Adaptor		Ansody - Portable Charger		YBBGANTE - Cable converter	
		Total	Weighted	Total	Weighted	Total	Weighted	Total	Weighted
Convenience	0.1	3	0.3	3	0.3	3	0.3	4	0.4
Power storage	0.2	4	0.8	0	0	4	0.8	0	0
cable management	0.2	4	0.8	3	0.6	3	0.6	4	0.8
Supports many cable types	0.2	4	0.8	3	0.6	2	0.4	3	0.6
easy of use	0.05	2	0.1	4	0.2	4	0.2	4	0.2
Quality	0.05	3	0.15	3	0.15	3	0.15	2	0.1
Price	0.2	2	0.4	3	0.6	3	0.6	4	0.8
Total Score	1		3.4		2.45		3.05		2.9

2. Brief Commentary on Success Factors

Convenience is the weight and size of the product because it is supposed to be easy to transport. Power storage for the traveling ability. Cable management to keep everything organized to keep it portable. Support for multiple cable types allows for a range of devices to be used. Quality specifically looks at how sturdy the product is. The cheaper the price is the higher it is scored.

3. Competitor Descriptions

Vintar is a company that has created a universal travel adaptor that contains many male plugs for easy to travel charging connections. It has a sliding mechanism that releases the desired male plug and a button mechanism that retracts it. It is a compact size of 2.1" x 2.1" x 2.1" and 5.6 oz. The charging outlets contain 2 usb-A 3 usb-C and AC outlets. This allows for any kind of traveler to be prepared for their charging needs in a stationary location.

The Ansody company creates a portable charger that contains a 25,000mAh battery pack. The size is 6.48" x 3.19" x 1.08" with a weight of 8.46 oz. It has 2 usb-C ports and a usb-A port allowing up to 3 devices to be charged at once. It is also compatible charging for macbook pro. It also provides usb-c to usb-c cable that is over 3ft long for the convenience of not having to purchase a separate one. Customers

purchase this product due to the amount of charge they have while on the go.

YBBGANTE company creates a cable converter. This allows for one type of cable to be changed to a usb-c type. While the output cable is limited in types the input cable has 7 different types it can convert from. Its size is 5" x 0.4" x 1.1" and weighs about 2 oz allowing easy travel. It has a rapid data transfer rate too that catches the eye of the consumer.

4. Score Justification (2-3 paragraphs total)

Convenience of each product was determined by the size and weight of the item if it was significantly large to hold the score was lower. The Vintar and YBBGANTE products both were small in size while the YBBGANTE was the lightest in weight. Both Ansody and our product were on the larger and heavy side due to the fact a battery is a dense item adding more weight to the products. Power storage is non-existent in the Vintar and YBBGANTE product giving them both 0 while the Ansody and ours has a 25000mAh battery storage. Ansody provides a usb-c to usb-c charger, where Vintar does not provide any cables but provides male adaptors to outlets and YBBGANTE provides an adaptor cable. Our product provides male adaptors and places to store them as all cables are embedded into the device.

Due to there being many different types of plug types not only just international power conversion but also individual cable adaptors needed our product provides many different connections to provide a solution. YBBGANTE also provides many different individual cable adaptors not power conversion adaptors though, Vintar is on the other side where it provides the international power conversion but not the individual cable adaptors, and Ansody does not provide any adaptors. Easy to use YBBGANTE is just a plug and use, no buttons to press, while Vintar has a few buttons and switches to push as well as Ansody. Our product has a bit more going on in the sense of items so a quick guide would be

useful. Quality of each product the YBBGANTE is very thin and easily breakable, where as everyone else has more strength but will still get a dent if thrown. From cheapest to most expensive YBBGANTE, Vintar, where our product can be just as expensive as Ansody. In the end our product beat out the competition by 0.35 because of our product containing every aspect it increased our value.